

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 3
 Duration: 25 minutes

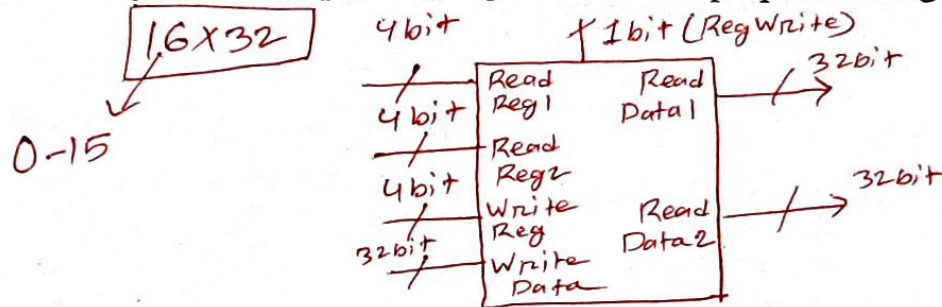
Semester: Summer 2025
 Full Marks: 15

CSE 340: Computer Architecture

Name:	ID:	Section:
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1. Suppose we are working with a 32bit version of RISC-V where we have 16 registers. Now design the corresponding register file with proper labeling [3]

Answer:



2. True/False. In case a statement is False, provide the correct statement.

- a) In a single-cycle RISC-V datapath, the store instruction takes the longest time to execute compared to other instructions. [1]
- b) Forwarding completely eliminates all stalls in a pipelined RISC-V processor. [1]
- c) The ALUOp signal alone is sufficient to uniquely determine all ALU operations in a RISC-V datapath. [1]
- d) In a pipelined datapath, both instruction fetch and data memory access can happen in the same clock cycle. [1]
- e) In the RISC-V pipelined datapath, separate instruction and data memories are required to prevent structural hazards. [1]

Answer:

- a) False. Load takes highest time.
- b) False. $\text{Ld } x_2, 16(x_5)$
 $\text{Add } x_5, x_2, x_3$ } only forward can't help here. I stall is a must.
- c) False. Need the help of "ALU Control".
- d) True.
- e) True.

3.

add X1, X2, X7

sub X2, X1, X5

sd X2, 16(X1)

ld X2, 16(X5)

sll X5, X6, X7

add X2, X1, X7

The above mentioned instruction is running on a pipelined datapath. Answer the following questions based on this.

a) Solve the data hazard using only stalling. [4]

	CC1	CC2	CC3	CC4	CC5	CC6	CC7	CC8	CC9	CC10	CC11	CC12	CC13	CC14
(i)	IF	ID	EX	MEM	WB									
		X	X	X	X	X								
			X	X	X	X	X							
(ii)				IF	ID	EX	MEM	WB						
				X	X	X	X	X						
					X	X	X	X	X					
(iii)						IF	ID	EX	MEM	WB				
(iv)							IF	ID	EX	MEM	WB			
(v)								IF	ID	EX	MEM	WB		
(vi)									IF	ID	EX	MEM	WB	

b) How many clock cycles are needed to run the instruction set? [1]

Answer: 14

Instruction Fetch takes 15ps

Register read/write takes 15ps

ALU operation takes 30 ps

Memory access takes 15ps

c) Find the clock period. [1]

Answer: 30ps

d) Total time to run the instruction. [1]

Answer: $(14 \times 30) = 420ps$